

EKELE OGBADU

Lead AI/ML Engineer | CYBERCOM J7, Agentic AI, RAG, LLM Systems, Secure AI Platforms
Ellicott City, MD • [linkedin.com/in/ekele-ogbadu](https://www.linkedin.com/in/ekele-ogbadu)

SUMMARY

Lead AI/ML Engineer with 12+ years of software engineering experience and 4+ years of applied machine learning experience supporting mission-critical defense, cyber, and AI modernization programs. Currently serving as the primary AI/ML engineer for U.S. Cyber Command J7, leading directorate-level AI/ML initiatives across LLM-enabled workflows, semantic retrieval, tagging, data pipelines, evaluation systems, and operational AI adoption. Ph.D. candidate in Computer Science specializing in multimodal AI, grounded language understanding, and human-robot interaction, with hands-on experience across NLP, retrieval, computer vision, sensor fusion, and production ML systems.

PROFESSIONAL EXPERIENCE

Booz Allen Hamilton — Annapolis Junction, MD

Lead AI/ML Engineer | U.S. Cyber Command J7 Support | June 2026 – Present

- Serve as a lead AI/ML engineer supporting U.S. Cyber Command J7, leading AI/ML training, tool development, platform enablement, and secure transition of AI capabilities into mission environments.
- Conduct AI/ML training for J7 personnel and contribute to enterprise training approaches that standardize how Cyber Command users learn, evaluate, and apply AI-enabled tools.
- Develop AI-enabled tools for the AI Gallery, translating mission needs into reusable capabilities for cyber training, knowledge discovery, experimentation, and operational support.
- Build AI platform tools and agentic workflows using BASE-C, LangChain/LangGraph patterns, and MCP-style tool integration to support secure tool access, workflow orchestration, and transition of AI capabilities into high-side mission environments.
- Design LLM-enabled and RAG-style workflows for tagging, semantic retrieval, ontology-driven classification, vector search, and AI-assisted knowledge discovery using Python, embeddings, AWS Bedrock, OpenSearch, and evaluation pipelines.
- Create gold set generation workflows, retrieval evaluation loops, prompt evaluation methods, and metric-driven refinement processes to improve tagging quality, retrieval relevance, and trust in AI-enabled mission tools.
- Partner with J7 stakeholders, cyber mission users, engineers, and platform teams to identify AI use cases, define implementation paths, and deliver secure AI/ML capabilities aligned to command-level needs.

Senior AI/ML Engineer | Sep 2025 – May 2026

- Delivered the team's tagging pipeline end to end, completing a long-standing capability after rapidly learning a large codebase, designing the solution, and integrating it into the broader application.
- Designed and implemented an ontology-driven tagging architecture that structured how unstructured content was classified, tagged, and surfaced across the broader application.
- Developed automated data ingestion, enrichment, and feature pipelines in Python to process large-scale document and signal collections.
- Architected RAG-style tagging and semantic retrieval workflows using embeddings, AWS Bedrock, OpenSearch, vector search patterns, and LLM-enabled ranking to support scalable indexing, querying, and discovery of unstructured data.
- Built multi-step AI workflows connecting ingestion, enrichment, tagging, retrieval, evaluation, and user-facing AI capabilities.
- Built gold set generation workflows and MLOps-aligned evaluation loops to improve tagging accuracy, retrieval relevance, model quality, and ongoing system performance.
- Partnered across engineering and product teams to translate mission needs into reliable, production-ready AI/ML capabilities.

Senior Agile Engineer (ML Programs) | Jul 2024 – Sep 2025

- Led release operations and technical delivery coordination across multiple teams on a \$200M mission program, serving as a central point of alignment for stakeholders, dependencies, and execution risk.
 - Drove delivery improvements that reduced average days to completion from 167.3 to 36 and cut blocked-release time from 250.5 days to 0.
 - Led daily scrums across four sites, facilitated weekly classified stakeholder meetings, and presented release metrics, bottlenecks, and improvement opportunities during PI Planning and other program forums.
 - Improved delivery predictability and stakeholder confidence through disciplined execution, issue resolution, and cross-team coordination.
-

University of Maryland, Baltimore County — Interactive Robotics & Learning Lab

AI/ML Graduate Research Assistant | May 2023 – Present

- Conducted applied research in multimodal grounding and grounded language understanding for human-robot interaction, integrating text, images, and sensor data including 3D point clouds, audio, and ROS bag files.
 - Developed SCOUT++ from the original SCOUT dataset by cleaning, restructuring, and aligning multimodal data into 11,000+ timestamp-aligned instruction-image pairs for grounded instruction understanding and multimodal learning experiments.
 - Designed and implemented end-to-end ML pipelines for preprocessing, synchronization, model training, and evaluation across multimodal datasets using PyTorch, TensorFlow, Pandas, NumPy, OpenCV, and ROS.
 - Evaluated neural and GPT-based approaches for grounded instruction understanding, including work on ambiguity handling, contextual reasoning, and clarification behavior in robot interaction.
 - First author on peer-reviewed research presented at the AAAI 2025 Fall Symposium with additional first author work being presented at the 2026 IEEE RO-MAN Conference.
-

Prescient Edge Corporation — McLean, VA

Software Engineering Team Lead | Aug 2023 – Jul 2024

- Led engineers maintaining the MINOTAUR platform across C++, Java, React, and TypeScript, providing technical direction, code review standards, onboarding support, and delivery oversight Improving junior engineer ramp-up time by 60%.

Machine Learning Engineer | May 2022 – Aug 2023

- Developed object detection and computer vision capabilities for Hull Identification Number recognition using YOLO, TensorFlow, and OpenCV across infrared and visible imagery.
- Built real-time sensor-fusion pipelines combining radar and camera inputs to track high-speed vessels and automate camera tilt and zoom.

Software Developer | Feb 2021 – May 2022

- Built and maintained mission-critical software deployed on U.S. Coast Guard rescue aircraft, integrating C++, Java, CI/CD, UDP/TCP, MIL-STD-1553, ARINC-429, and simulation tooling for hardware–software validation.
-

United States Navy

Aviation Administrationman | Oct 2014 – Oct 2018

- Supervised night-shift aviation maintenance administration, managed 900+ physical and digital records, and used SQL/enterprise systems including NALCOMIS OMA and SYBASE to track equipment lifecycle, readiness data, maintenance schedules, and personnel trends.
 - As the official VAQ-132 Command Webmaster developed, and maintained the VAQ-132 command website.
-

Early Entrepreneurial Experience

Founder / Independent Software Developer | 2007 – 2018

- Built and launched web platforms, mobile applications, games, matching platforms, and business-driven software products across iOS, Android, and the web, including ventures through Born Royal LLC.
-

EDUCATION

University of Maryland, Baltimore County

- Ph.D. Candidate, Computer Science — Expected Dec 2027
Research: Multimodal NLP and perception for real-time human-robot interaction
- M.S., Computer Science — May 2024
- B.S., Computer Science (Data Science concentration) — May 2022
- B.A., Mathematics — May 2022

Edinboro University of Pennsylvania

Undergraduate Coursework in Computer Science — 2003 – 2007

Completed 110 credits toward a B.S. in CS before leaving to build & launch entrepreneurial software products.

TECHNICAL SKILLS

Languages: Python, C++, Rust, Java, JavaScript, TypeScript, SQL, Bash

AI/ML & Deep Learning: Machine Learning, Deep Learning, NLP, Computer Vision, Multimodal AI, Information Retrieval, Semantic Search, Vector Search, Retrieval-Augmented Generation (RAG), Embeddings, Classification, Object Detection, Sensor Fusion, Grounded Language Understanding, Human-Robot Interaction, PyTorch, TensorFlow, scikit-learn, OpenCV

LLM & Agentic AI Systems: LLM Applications, AI Agents, Multi-Step AI Workflows, Tool-Using Agents, Agentic AI, LangChain, LangGraph, MCP Servers, Prompt Engineering, Prompt Evaluation, AI Workflow Orchestration, LLM Tool Integration, RAG Pipelines, Knowledge Discovery Systems

MLOps, Evaluation & Production AI: Model Evaluation, Retrieval Evaluation, Gold Set Generation, Model Monitoring, Experiment Tracking, Dataset Versioning, ML Pipelines, CI/CD for ML Systems, Production ML Workflows, Model Quality Assessment, Human-in-the-Loop Evaluation, Responsible AI, AI Safety, AI Governance

Data, Search & Platforms: OpenSearch, Vector Databases, PostgreSQL, MongoDB, Pandas, NumPy, Large-Scale Data Processing, Data Ingestion, Data Enrichment, Ontology-Driven Classification, Metadata Extraction

Cloud, Infrastructure & Secure Platforms: AWS, AWS Bedrock, EC2, S3, SageMaker, BASE-C, Docker, Kubernetes, REST APIs, Microservices, CI/CD, Secure AI Platform Development, High-Side AI Transition, ROS

Software Engineering & Leadership: System Design, AI Platform Engineering, Technical Leadership, Cross-Functional Delivery, Agile/Scrum, Release Management, Stakeholder Engagement, Mission Engineering, Startup Product Development, Open-Source AI Projects, Git, GitHub/GitLab, Jupyter, React, Node.js

CERTIFICATIONS

AWS Certified Machine Learning – Specialty | AWS Certified Cloud Practitioner | CompTIA Security+ ce | Certified SAFe® 6 Practitioner | Artificial Intelligence Engineer Expert — Booz Allen Hamilton | Artificial Intelligence Engineer Practitioner — Booz Allen Hamilton

PUBLICATIONS

- Ekele Ogbadu, Stephanie Lukin, and Cynthia Matuszek, *Grounded Instruction Understanding with Large Language Models: Toward Trustworthy Human-Robot Interaction*. **Published in Proceedings of the AAI Symposium Series, 7(1), 223–231, 2025.**
- Ekele Ogbadu, *Ask-to-Act: Learning When Robots Should Clarify Ambiguous Grounded Instructions*. **Accepted to IEEE RO-MAN 2026.**